Digital Input MAPS Template (Instrumentation)

The Digital Input MAPS template is designed to control a digital input.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This MAPS template is available in three models – advanced, standard and basic, which provide respectively lower levels of control over the digital input.

This document describes the Advanced Digital Input MAPS Template and also defines the level of functionality provided by the standard and basic models. So that by reading this document you can decide which model best suits your needs. You should consider the following factors in your choice:

- the importance of the digital being controlled and
- the budget of the customer

Since the more complex the model, the greater the required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Advanced Digital Input Functional Philosophy

The digital input is sent to the control program once the field input is triggered or an operator enables it from the faceplate.

Field Inputs

- Digital Input

Simulation mode - will override the input to the digital input block and simulate a digital input in the PLC program that is controlled from the faceplate. Simulation mode is used to test the process logic without receiving the physical input itself. It is very useful in testing, commissioning and maintenance.

Typical Faceplate Graphics for the Advanced Digital Input MAPS Template

Digital Input	
	
	Control/ Tag name
	Simulation mode indication
	Digital input status indication

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Control Words:					
SSCL 0	Control Word	**Marshal**	**x**	**x**	**x**
SSCL 1	Delay On Setpoint Time	**Analog**	**x**	**x**	
SSCL 2	Delay Off Setpoint Time	**Analog**	**x**		

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Status Words:					
SSSL 0	Status Word	**Marshal**	x	x	x

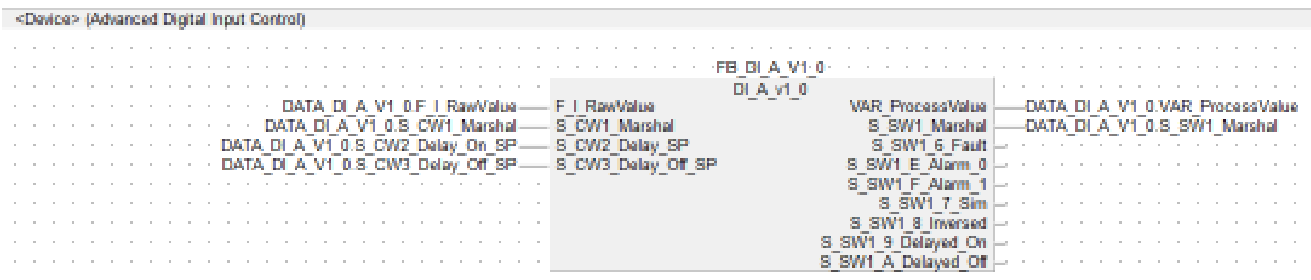
Digital Input:				
DI 0	Digital Input	x	x	x

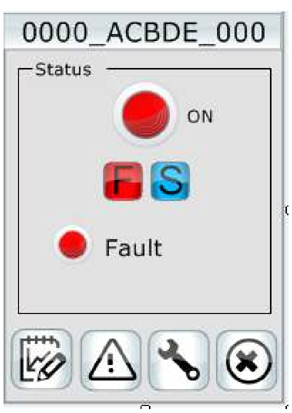
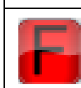
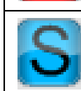
SCADA Control Word (Adroit Marshal Agent)		Event Logged	Advanced	Standard	Basic
Bit 0	Input inverted	x	x	x	x
Bit 1	Spare				
Bit 2	Spare				
Bit 3	Simulation Enabled	x	x	x	x
Bit 4	Simulation Value	x	x	x	x
Bit 5	Alarm Enabled	x	x	x	x
Bit 6	Alarm on bit off	x	x	x	x
Bit 7	Alarm on bit on	x	x	x	x
Bit 8	Spare				
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

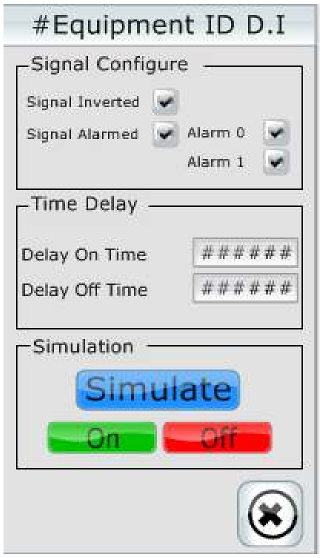
SCADA Status Word (Adroit Marshal Agent)		Alarmed	Advanced	Standard	Basic
Bit 0	Spare				
Bit 1	Spare				
Bit 2	Input Status Process Value		x	x	x
Bit 3	Spare				
Bit 4	Spare				
Bit 5	Spare				
Bit 6	Fault Active	x	x	x	x
Bit 7	Simulation Enabled		x	x	x
Bit 8	Input Inverted		x	x	x
Bit 9	Delayed On		x	x	
Bit 10	Delayed Off		x		
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Alarm bit 0	x	x	x	x
Bit 15	Alarm bit 1	x	x	x	x

Resources	Advanced	Standard	Basic
Digital Inputs	1	1	1
Digital Outputs	0	0	0
Analog Inputs	0	0	0
Analog Outputs	0	0	0
PLC Memory Steps	159	122	108
System Words Used	4	1	0
System Bits Used	64	59	58
SCADA SCAN Points	4	3	2

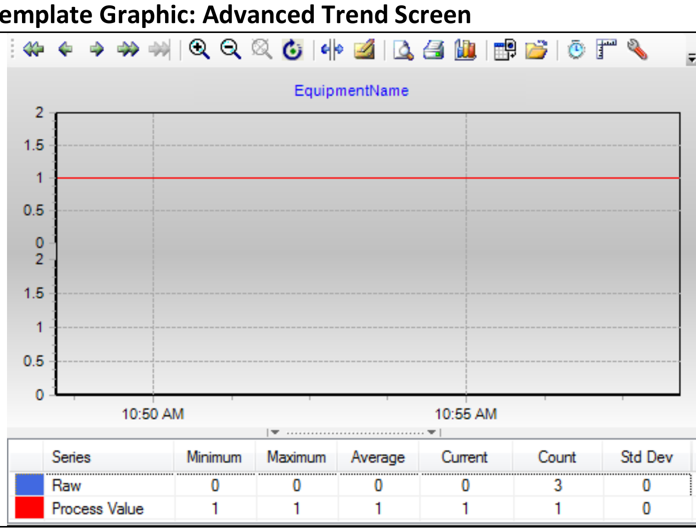
Template: Advanced Digital Input PLC Function Block



Template Graphic: Advanced Home Screen	Control	Descriptions
		Digital Input controlled name
		High-high alarm indication
		High alarm indication
		Low alarm indication
		Low-low alarm indication

<table>
<tr><th>Template Graphic: Advanced Setup Screen</th><th>Control</th><th>Descriptions</th></tr>
<tr><td rowspan="10">
</td><td>#Equipment ID</td><td>Digital Input Name</td></tr>
<tr><td>Signal Inverted</td><td>Enable the digital input signal to be inverted</td></tr>
<tr><td>Signal Alarmed</td><td>Enable the digital input signal to be alarmed</td></tr>
<tr><td>Alarm 0</td><td>Alarm on bit off</td></tr>
<tr><td>Alarm 1</td><td>Alarm on bit on</td></tr>
<tr><td>Delay On Time</td><td>Delay the digital input to switch on</td></tr>
<tr><td>Delay Off Time</td><td>Delay the digital input to switch off</td></tr>
<tr><td>Simulate</td><td>Simulate the digital input</td></tr>
<tr><td>On</td><td>Simulate the digital input on</td></tr>
<tr><td>Off</td><td>Simulate the digital input off</td></tr>
</table>

<table>
<tr><th>Template Graphic: Advanced Alarms & Events Screen</th><th>Control</th><th>Descriptions</th></tr>
<tr><td rowspan="4">
</td><td>Alarms</td><td>Display all the alarms filtered for the current Digital Output</td></tr>
<tr><td>Events</td><td>Display all the events filtered for the current Digital Output.</td></tr>
<tr><td>Local Acknowledge</td><td>Acknowledge all the alarms on locally on the current machine</td></tr>
<tr><td>Global Acknowledge</td><td>Acknowledge all the alarms on globally on all the machines.</td></tr>
</table>

<table>
<tr><th>Template Graphic: Advanced Trend Screen</th><th>Control</th><th>Descriptions</th></tr>
<tr><td rowspan="3">
</td><td>Equipment Name</td><td>Display the name of the current digital Input of the trend</td></tr>
<tr><td>Raw</td><td>Display the raw digital input value</td></tr>
<tr><td>Process Value</td><td>Display the processed digital input value</td></tr>
</table>